

Baishen Liang

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EDUCATION

<i>Doctor of Philosophy in Cognitive Neuroscience</i> University of Chinese Academy of Sciences, Beijing, China	2017 – 2023 GPA: 3.94
<i>Exchanged Student, Department of Linguistics</i> University of California, Berkeley, Berkeley, CA, USA	2016 – 2016 GPA: 3.78
<i>Bachelor of Science in Psychology</i> Central China Normal University, Wuhan, China	2013–2017 GPA: 3.88

WORK EXPERIENCE

Postdoctoral Associate Department of Neurology, Duke University, Durham, NC, USA	July 2024 – present
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- Lead a project to Design and establish an experiment to **study production-based verbal working memory maintenance and manipulation**. Collect, analyze, and maintain neurosurgical patients' intracranial electroencephalogram (iEEG) data.

PhD Student Institute of Psychology, Chinese Academy of Sciences, Beijing, China	September 2017 – June 2023
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Technology Consultant for the Core Facility IPCAS (Nominated Position)

- Assisted in maintaining transcranial magnetic stimulation (TMS) facilities.
- Gave regular public lectures and courses for faculty and students.
- Provided one-on-one technical consultation for faculty and students requiring TMS.

Project 1: Bilateral Laryngeal Motor Cortex in Speech Perception in Noise

- Lead a project to apply functional magnetic resonance imaging (fMRI) to localize Mandarin speakers' laryngeal motor cortices. Designed and established a TMS experiment, and collected and analyzed data.

Project 2: Transcranial Alternative Electric Stimulation (tACS) on Auditory and Motor Cortex

- Led a project to establish a technical pipeline combining phase-controlled dual-site tACS and electroencephalogram (EEG) recordings for auditory cognitive neuroscience research.

Project 3: Cortical Functional Map for the Perception of Speech Features

- Systematically reviewed 148 published fMRI studies on speech perception and comprehension.

PEER REVIEWED PUBLICATIONS

1. **Baishen Liang**, Yanchang Li, Wanying Zhao, Yi Du*. Bilateral human laryngeal motor cortex in perceptual decision of lexical tone and voicing of consonant. *Nature Communications*, 2023, 14, 4710. (First author, **Journal Impact Factor = 14.7**)
2. **Baishen Liang** & Yi Du*. The functional neuroanatomy of lexical tone perception: an activation likelihood estimation meta-analysis. *Frontiers in Neuroscience*, 2018, 12, 495. (First author, **cited by 62 publications**)
3. **Baishen Liang** & Yi Du*. Mechanisms of sensorimotor integration in speech perception [言语知觉中的感觉运动整合机制]. *Science & Technology Review* [科技导报], 2017, 35(19):21-28. (First author)

MANUSCRIPTS IN PROGRESS

1. **Baishen Liang**, A. M. Earle-Richardson, G. Grant, M. Zafar, B. Frauscher, D. Southwell, G. Hickok, G. B. Cogan*. Sensory-motor mechanisms for verbal working memory. In preparation. (First author)
2. Jiali Liu[#], Yixiang Wang[#], Jiayu Feng[#], Haichao Zhao, Qinghua He, Xuefeng Yang*, **Baishen Liang***. Structural learning of syllable sequences. Under review. (Co-corresponding author)
3. Yaxuan Wang[#], Keke Yu[#], Shuqi Yin, **Baishen Liang***, Ruiming Wang*. Theta and gamma transcranial alternating current stimulation modulate Mandarin consonant and lexical tone perception. *bioRxiv*, 2025, 679546, under review. (Co-corresponding author)
4. Weijin Hu[#], **Baishen Liang**[#], Yi Du*, Singing enhances poetry learning and memorization in preschool children. Under review. (Co-first author)

HONORS & AWARDS

- **Best Presentation Award in the 4th IEEE EMBS International Summer School of Neural Engineering**. Tsinghua University, Beijing, August 2018.
- **University of Chinese Academy of Sciences Student Grant (¥10,000)**. University of Chinese Academy of Sciences, Beijing. December 2017.
- **National Scholarship (¥8,000)**. Ministry of Education of the People's Republic of China, Beijing. November 2016.

PROFESSIONAL MEETINGS ATTENDED

- **Liang, B.**, Earle-Richardson, A. M., Grant, G., Zafar, M., Frauscher, B., Southwell, D., Hickok, G. & Cogan, G. B.* Sensory-motor mechanisms for verbal working memory.
 - *Neuroscience 2025*, Society for Neuroscience. San Diego, CA, USA, November 2025 (poster).
 - *Advances and Perspectives in Auditory Neuroscience*. San Diego, CA, USA, November 2025 (poster).

- *Society for the Neurobiology of Language 17th Annual Meeting*. Washington DC, USA, September 2025 (poster).
- Duke Comprehensive Epilepsy Center Research Colloquium. Durham, USA, September 2025 (slide presentation)
- *Duke Neurology TBS/DCEC Research Symposium*. Durham, USA, September 2025 (poster)
- *North Carolina Conference on Cognition 2025*. Chapel Hill, USA, April 2025 (poster).
- **Liang, B.**, Li, Y., Zhao, W., Du, Y.* Bilateral human laryngeal motor cortex in the perceptual decision of lexical tone and voicing of consonant (poster and presentation). *Society for the Neurobiology of Language Annual Meeting 2023*. Marseille, France, October 2023.
- **Liang, B.**, Hong, Y., Li, Y., Du, Y.* Causal involvement of auditory-motor low-frequency oscillation coupling in sentence perception under noisy conditions: a tACS study (poster). *Society for the Neurobiology of Language Annual Meeting 2023*. Marseille, France, October 2023.
- **Liang, B.**, Li, Y., Zhao, W., Du, Y.* A causal role of the human laryngeal motor cortex in lexical tone and voicing perception (poster and presentation). *Neurobiology of Language: Key Issues and Ways Forward II*. Nijmegen, Netherlands (online), March 2022.
- **Liang, B.**, Du, Y.* Neuroanatomy of lexical tone: an activation likelihood estimation meta-analysis (poster). *Organization for Human Brain Mapping Annual Meeting 2018*. Singapore, June 2018.

PEER REVIEW EXPERIENCE

Nature, 2025

Brain and Language, 2025

International Journal of Multilingualism, 2025

Acta Psychologica Sinica [心理学报], 2024

TECHNICAL SKILLS

▪ Programming languages

- Matlab, Python, R, and C++.

▪ Psychology

- Psychophysics experimental designs.
- Statistics: linear models, logistic models, and drift diffusion models (DDMs).

▪ Linguistics

- Acoustic processing and feature extractions by Praat/Matlab.

▪ Neuroscience

- Electroencephalography (EEG), Magnetoencephalography (MEG), and intracranial EEG experimental design, data collection, preprocessing, and analyses.
- Transcranial magnetic stimulation (TMS) and transcranial electrical stimulation (tES) experimental design, data collection, and analyses.